D(z)-quadruples in Z[√-2]

Problèmes ouverts — A. Dujella · 2026

Problem 1.8 (Dujella, 2026). Characterize elements z of the ring $\mathbb{Z}[\sqrt{-2}]$ for which there exists a $D(z)$-quadruple.

Remark. Partial results on (non)existence of $D(z)$-quadruples in $\mathbb{Z}[\sqrt{-2}]$ were obtained by Abu Muriefah and Al-Rashed (Math. Commun. **9** (2004), 1–8), Franušić (Math. Commun. **9** (2004), 141–148), Dujella and Soldo (An. Ştiinţ. Univ. "Ovidius" Constanţa Ser. Mat. **18** (2010), 81–98), and Soldo (Miskolc Math. Notes **14** (2013), 265–277).